

1 nf-core/coproID v2.0: An improved pipeline for the 2 identification of (palaeo)faecal depositors

3 Meriam van Os ^{1,2} and Maxime Borry ^{2,3}

4 1 Department of Anatomy, University of Otago, Dunedin, New Zealand 2 Microbiome Sciences Group,
5 Department of Archaeogenetics, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany
6 3 Associated Research Group of Archaeogenetics, Leibniz Institute for Natural Product Research and
7 Infection Biology Hans Knöll Institute, Jena, Germany ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright
and release the work under a
Creative Commons Attribution 4.0
International License ([CC BY 4.0](#))

8 Summary

9 With the advancement of Next Generation Sequencing technologies, (palaeo)faeces have
10 become a unique and valuable source in the fields of archaeology ([Battillo, 2019](#)), microbiome
11 studies ([Rifkin et al., 2020](#); [Wibowo et al., 2021](#)), and species ecology and conservation ([Ang
12 et al., 2020](#); [Taylor et al., 2022](#)), and even shows promise in forensic investigations ([Frederike
13 C. A. Quaak et al., 2017](#); [Frederike C. A. Quaak et al., 2018](#)). To use faecal samples for such
14 studies, the central and/or first question is often “who deposited the faeces?”, before moving
15 onto further analyses. The nf-core/coproID pipeline helps to answer this question by taking
16 raw sequencing data and predicting the depositor’s species.

17 The raw sequencing data is first pre-processed by nf-core/coproID to trim adapters and remove
18 low quality and low complexity reads with fastp ([Chen et al., 2018](#)). Bowtie2 ([Langmead et
19 al., 2018](#)) is then used to align the reads to multiple customer specified reference genomes of
20 potential depositor (host) species. These reads are then processed with sam2lca ([Borry et al.,
21 2022](#)) to retain only reads specific to one of the references, and normalised according to the
22 size of the genome. Furthermore, a taxonomic profile is created with kraken2 ([Wood et al.,
23 2019](#)), and compared to a customer supplied database of potential sources using sourcepredict
24 to estimate the percentages of contributing sources ([Borry, 2019](#)). Both the normalised host
25 DNA and the sourcepredict results are used to predict the most likely depositor of the faeces.
26 The pipeline also incorporates ancient DNA damage estimates using pyDamage ([Borry et al.,
27 2021](#)) and damageprofiler ([Neukamm et al., 2021](#)) for authenticating the ancient nature of the
28 host DNA, when working with palaeofaeces. All results are collated into a Quarto notebook
29 html report for an easy overview of all the results.

30 Statement of need

31 As mentioned above, (palaeo)faeces are valuable resources to study the depositor’s DNA, diet,
32 microbiome, health and more. However, it is often difficult to distinguish between the depositor
33 based on the faeces morphology alone. For example, humans and dogs often overlap in their
34 diets, and produce similar faeces in size and shape. In 2020, the pipeline nf-core/coproID v1.0
35 was published ([Borry et al., 2020](#)), which uses both host and microbial DNA to predict the
36 depositor of faecal samples. The microbiome can be a crucial part for a host prediction, as the
37 host DNA content in faeces can be very low in certain species and/or individuals ([Ang et al.,
38 2020](#); [Perry et al., 2010](#)), including humans and modern dogs ([Borry et al., 2020](#)). Since its
39 first release, new tools have become available that can improve the accuracy and usability of
40 nf-core/coproID. Here we present the newest version of the pipeline, nf-core/coproID v2.0.0,
41 rewritten in the newest Nextflow DLS2 language and with newly incorporated tools.

Materials and Methods

nf-core/coproID is a bioinformatics pipeline that helps identify the “true depositor” of DNA whole genome shotgun sequenced (palaeo)faeces by analysing the microbiome composition together with the endogenous host DNA.

It combines the analysis of the putative host (ancient) DNA with a machine learning prediction of the faeces source, based on microbiome taxonomic composition:

1. First, coproID performs parallel mapping of all reads against two (or more) target genomes (genome1, genome2, ..., genomeX) using bowtie2 (Langmead et al., 2018), and computes a host-DNA species ratio (NormalisedProportion) using sam2lca (Borrry et al., 2022).
2. Next, coproID performs metagenomic taxonomic profiling with kraken2 (Wood et al., 2019), and compares the obtained profiles to user supplied modern reference samples of the target species metagenomes. Using machine learning, sourcepredict (Borrry, 2019) then estimates the host source from the metagenomic taxonomic composition (SourcepredictProportion).
3. Finally, coproID combines the A and B proportions to predict the likely host of the metagenomic sample.

Workflow

The newest version of nf-core/coproID, v2.0.0, was entirely rewritten in the newest DSL2 language of Nextflow to enhance modularity, reusability, and scalability (DI Tommaso et al., 2017). Additionally, various modifications were made to the workflow to improve accuracy and reporting.

Figure 1 describes the newest workflow:

1. Quality check of the input fastq reads with FastQC (Andrews, 2010).
2. Fastp is used to remove adapters and low-complexity reads (Chen et al., 2018).
3. Mapping of adapter trimmed reads to multiple reference genomes with Bowtie2 (Langmead et al., 2018).
4. Lowest Common Ancestor analysis with sam2lca (Borrry et al., 2022) to retain only genome specific reads, i.e. reads that align equally well to multiple references are identified as belonging to a Lower Common Ancestor and removed from the read counts. The sam2lca read counts are normalised by the size of the genome. First, a normalisation factor is calculated per reference, or source species (sp):

$$AverageReferenceLength = \sum_{sp} ReferenceLength_{sp} / NumberofReferences$$

$$NormalisationFactor_{sp} = AverageReferenceLength / ReferenceLength_{sp}$$

The normalised read counts were then calculated by:

$$NormalisedReads_{sp} = sam2lcaReads_{sp} * NormalisationFactor_{sp}$$

5. Taxonomic profiling is performed on adapter trimmed reads with kraken2 (Wood et al., 2019), and by using a custom supplied database. Kraken2 reports are then parsed and merged into one table for all samples.
6. Sourcepredict (Borrry, 2019) is then used to predict the source proportions, based on the kraken2 taxonomic profiles, and by using customer supplied reference sources (which should have been created with the same reference database).

80 7. Both the host DNA (NormalisedReads) and sourcepredict proportion are used to predict
81 the most likely depositor of the (palaeo)faeces. The probability of each reference species
82 is calculated by:

$$Probability_{sp} = NormalisedSam2lcaProportion_{sp} * SourcepredictProportion_{sp}$$

83 8. Ancient DNA damage patterns are estimated using pyDamage (Borry et al., 2021) and
84 damageprofiler (Neukamm et al., 2021) to assess the authenticity of the ancient nature
85 of the DNA for when customers work on palaeofaecal samples.
86 9. MultiQC (Ewels et al., 2016) aggregates results of several individual nf-core modules.
87 10. Quaronotebook (Allaire et al., 2024) creates a report with an overview of all sample
88 results (incl. tables and figures). This includes the (normalised) sam2lca results and
89 calculations, the sourcepredict results, and DNA damage patterns.

90 Output

91 The results are located in a nested folder architecture. Fourteen subfolders are created within
92 the customer identified output folder: - bowtie2 - create - damageprofiler - fastp - fastqc -
93 kraken - kraken2 - multiqc - pipeline_info - pydamage - quaronotebook - sam2lca - samtools -
94 sourcepredict

95 These subfolders contain the main outputs from the concerned analyses. When the kraken2
96 database is supplied as a archive file (*.tar.gz), and/or the sourcepredict supplied taxa_sqlite
97 file (ending in .xz), the additional folders - untar - xz are created with the decompressed
98 files/folders.

99 Discussion and conclusions

100 Here we present a new version of the nf-core/coproID pipeline, designed to identify the true
101 depositor of (palaeo)faeces. Written in Nextflow DSL2, and adhering to the latest nf-core
102 standards and guidelines, nf-core/coproID v2.0.0 is more modular, reusable, and scalable. It
103 includes several new features, including fastp for faster pre-processing of the sequencing reads,
104 sam2lca to improve and generalize host DNA prediction, pyDamage to discriminate between
105 ancient and modern DNA, and the automated creation of an overall Quarto notebook html
106 report. The modular design of nf-core/coproID v2.0.0 by using Nextflow's most recent DSL2
107 and nf-core modules, also makes it easier for users to customise the pipeline, for example by
108 adding more modules and workflows.

109 Funding source declaration

110 MO was supported by a University of Otago Doctoral Scholarship.

111 Availability

112 The nf-core/coproID pipeline is freely available from the nf-core pipelines depository <https://nf-co.re/coproID/2.0.0/>.
113

Figures

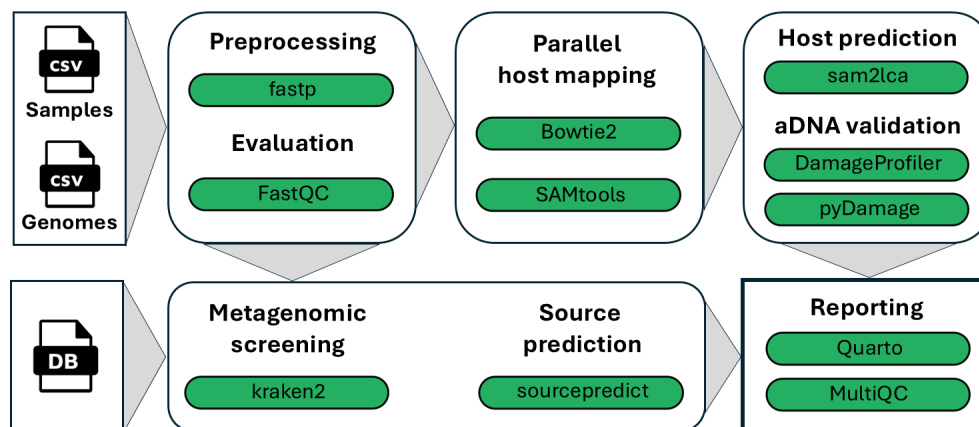


Figure 1: Figure 1

References

- Allaire, J. J., Teague, C., Scheidegger, C., Xie, Y., Dervieux, C., & Woodhull, G. (2024). *Quarto* (Version 1.6). <https://doi.org/10.5281/zenodo.5960048>
- Andrews, S. (2010). *Babraham bioinformatics - FastQC a quality control tool for high throughput sequence data*. <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>
- Ang, A., Roesma, D. I., Nijman, V., Meier, R., Srivathsan, A., & Rizaldi. (2020). Faecal DNA to the rescue: Shotgun sequencing of non-invasive samples reveals two subspecies of southeast asian primates to be critically endangered species. *Scientific Reports*, 10(1), 9396–9396.
- Battillo, J. (2019). Farmers who forage: Interpreting paleofecal evidence of wild resource use by early corn farmers in the north american southwest. *Archaeological and Anthropological Sciences*, 11(11), 5999–6016.
- Borri, M. (2019). Sourcepredict: Prediction of metagenomic sample sources using dimension reduction followed by machine learning classification. *Journal of Open Source Software*, 4(41), 1540. <https://doi.org/10.21105/joss.01540>
- Borri, M., Cordova, B., Perri, A., Wibowo, M., Prasad Honap, T., Ko, J., Yu, J., Britton, K., Girdland-Flink, L., Power, R. C., Stuijts, I., Salazar-García, D. C., Hofman, C., Hagan, R., Samdapawindé Kagoné, T., Meda, N., Carabin, H., Jacobson, D., Reinhard, K., ... Warinner, C. (2020). CoproID predicts the source of coprolites and paleofeces using microbiome composition and host DNA content. *PeerJ (San Francisco, CA)*, 8, e9001–e9001.
- Borri, M., Hübner, A., Rohrlach, A. B., & Warinner, C. (2021). PyDamage: Automated ancient damage identification and estimation for contigs in ancient DNA de novo assembly. *PeerJ*, 9, e11845. <https://doi.org/10.7717/peerj.11845>
- Borri, M., Hübner, A., & Warinner, C. (2022). sam2lca: Lowest common ancestor for SAM/BAM/CRAM alignment files. *Journal of Open Source Software*, 7(74), 4360. <https://doi.org/10.21105/joss.04360>
- Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17), i884–i890. <https://doi.org/10.1093/BIOINFORMATICS/>

143 [BTY560](#)

- 144 DI Tommaso, P., Chatzou, M., Floden, E. W., Barja, P. P., Palumbo, E., & Notredame, C.
 145 (2017). Nextflow enables reproducible computational workflows. *Nature Biotechnology*
 146 2017 35:4, 35(4), 316–319. <https://doi.org/10.1038/nbt.3820>
- 147 Ewels, P., Magnusson, M., Lundin, S., & Käller, M. (2016). MultiQC: Summarize analysis
 148 results for multiple tools and samples in a single report. *Bioinformatics*, 32(19), 3047–3048.
 149 <https://doi.org/10.1093/bioinformatics/btw354>
- 150 Langmead, B., Wilks, C., Antonescu, V., & Charles, R. (2018). Scaling read aligners to
 151 hundreds of threads on general-purpose processors. *Bioinformatics*, 35(3), 421–432.
 152 <https://doi.org/10.1093/bioinformatics/bty648>
- 153 Neukamm, J., Peltzer, A., & Nieselt, K. (2021). DamageProfiler: Fast damage pattern
 154 calculation for ancient DNA. *Bioinformatics*, 37(20), 3652–3653. <https://doi.org/10.1093/bioinformatics/btab190>
- 155
- 156 Perry, G. H., Marioni, J. C., Melsted, P., & Gilad, Y. (2010). Genomic-scale capture and
 157 sequencing of endogenous DNA from feces. *Molecular Ecology*, 19(24), 5332–5344.
- 158 Quaak, Frederike C. A., Graaf, M.-L. M. de, Weterings, R., & Kuiper, I. (2017). Microbial
 159 population analysis improves the evidential value of faecal traces in forensic investigations.
 160 *International Journal of Legal Medicine*, 131(1), 45–51.
- 161 Quaak, Frederike C. A., Wal, Y. van de, Maaskant-van Wijk, P. A., & Kuiper, I. (2018).
 162 Combining human STR and microbial population profiling: Two case reports. *Forensic*
 163 *Science International : Genetics*, 37, 196–199.
- 164 Rifkin, R. F., Vikram, S., Ramond, J.-B., Rey-Iglesia, A., Brand, T. B., Porraz, G., Val, A.,
 165 Hall, G., Woodborne, S., Le Bailly, M., Potgieter, M., Underdown, S. J., Koopman, J.
 166 E., Cowan, D. A., Van de Peer, Y., Willerslev, E., & Hansen, A. J. (2020). Multi-proxy
 167 analyses of a mid-15th century middle iron age bantu-speaker palaeo-faecal specimen
 168 elucidates the configuration of the “ancestral” sub-saharan african intestinal microbiome.
 169 *Microbiome*, 8(1), 62–23.
- 170 Taylor, R. S., Manseau, M., Redquest, B., Keobouasone, S., Gagné, P., Martineau, C., &
 171 Wilson, P. J. (2022). Whole genome sequences from non-invasively collected caribou faecal
 172 samples. *Conservation Genetics Resources*, 14(1), 53–68.
- 173 Wibowo, M. C., Yang, Z., Borry, M., Hübner, A., Huang, K. D., Tierney, B. T., Zimmerman,
 174 S., Barajas-Olmos, F., Contreras-Cubas, C., García-Ortiz, H., Martínez-Hernández, A.,
 175 Luber, J. M., Kirstahler, P., Blohm, T., Smiley, F. E., Arnold, R., Ballal, S. A., Pamp, S.
 176 J., Russ, J., ... Kostic, A. D. (2021). Reconstruction of ancient microbial genomes from
 177 the human gut. *Nature (London)*, 594(7862), 234–239.
- 178 Wood, D. E., Lu, J., & Langmead, B. (2019). Improved metagenomic analysis with Kraken 2.
 179 *Genome Biology*, 20(1), 1–13. <https://doi.org/10.1186/S13059-019-1891-0>